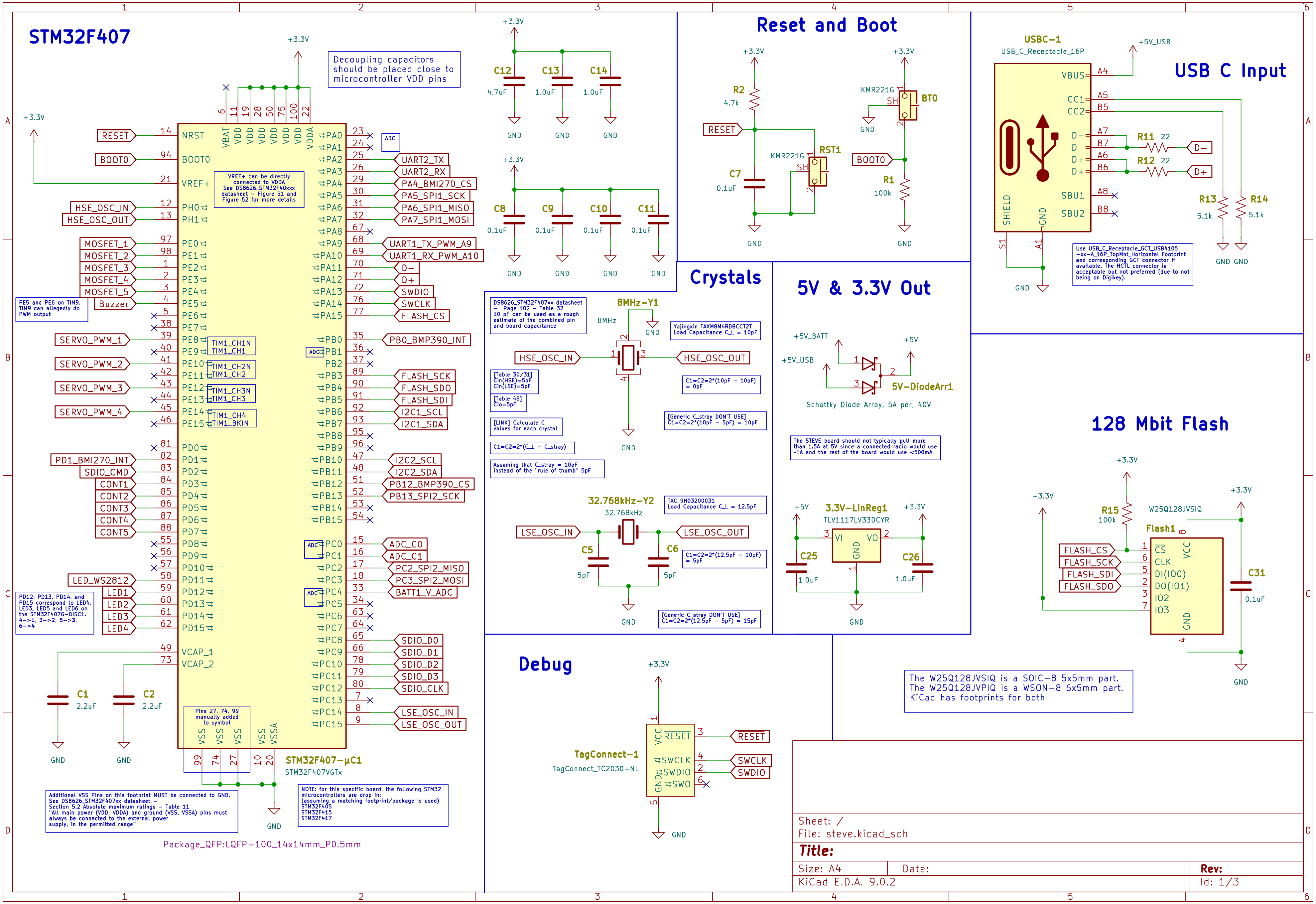

STM32F407
+3.3V
Decoupling capacitors should be placed close to microcontroller VDD pins
NRST
BOOT0
VREF+
VREF+ can be directly connected to VDDA See DS8626_STM32F40xxx datasheet – Figure 51 and Figure 52 for more details
RESET
BOOT0
HSE_OSC_IN
HSE_OSC_OUT
MOSFET_1
MOSFET_2
MOSFET_3
MOSFET_4
MOSFET_5
Buzzer
PE5 and PE6 on TIM9. TIM9 can allegedly do PWM output
SERVO_PWM_1
SERVO_PWM_2
SERVO_PWM_3
SERVO_PWM_4
TIM1_CH1N
TIM1_CH1
TIM1_CH2N
TIM1_CH2
TIM1_CH3N
TIM1_CH3
TIM1_CH4
TIM1_BKIN
PD1_BMI270_INT
SDIO_CMD
CONT1
CONT2
CONT3
CONT4
CONT5
LED_WS2812
LED1
LED2
LED3
LED4
PD12, PD13, PD14, and PD15 correspond to LED4, LED3, LED5 and LED6 on the STM32F407G-DISC1. 4->1, 3->2, 5->3, 6->4
VCAP_1
VCAP_2
C1 2.2uF
C2 2.2uF
GND
Pins 27, 74, 99 manually added to symbol
VSS VSS VSS VSS VSSA
STM32F407-µC1
STM32F407VGTx
Additional VSS Pins on this footprint MUST be connected to GND. See DS8626_STM32F407xx datasheet – Section 5.2 Absolute maximum ratings – Table 11 "All main power (VDD, VDDA) and ground (VSS, VSSA) pins must always be connected to the external power supply, in the permitted range"
NOTE: for this specific board, the following STM32 microcontrollers are drop in: (assuming a matching footprint/package is used) STM32F405 STM32F415 STM32F417
Package_QFP:LQFP-100_14x14mm_P0.5mm
ADC
UART2_TX
UART2_RX
PA4_BMI270_CS
PA5_SPI1_SCK
PA6_SPI1_MISO
PA7_SPI1_MOSI
UART1_TX_PWM_A9
UART1_RX_PWM_A10
D-
D+
SWDIO
SWCLK
FLASH_CS
PB0_BMP390_INT
FLASH_SCK
FLASH_SDO
FLASH_SDI
I2C1_SCL
I2C1_SDA
I2C2_SCL
I2C2_SDA
PB12_BMP390_CS
PB13_SPI2_SCK
ADC_C0
ADC_C1
PC2_SPI2_MISO
PC3_SPI2_MOSI
BATT1_V_ADC
SDIO_D0
SDIO_D1
SDIO_D2
SDIO_D3
SDIO_CLK
LSE_OSC_IN
LSE_OSC_OUT
C12 4.7uF
C13 1.0uF
C14 1.0uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
Reset and Boot
R2 4.7k
C7 0.1uF
KMR221G
RST1
BT0
R1 100k
Crystals
DS8626_STM32F407xx datasheet – Page 102 – Table 32 10 pF can be used as a rough estimate of the combined pin and board capacitance
8MHz-Y1
8MHz
Yajingxin TAXM8M4RDBCCT2T Load Capacitance C_L = 10pF
[Table 30/31] Cin(HSE)=5pF Cin(LSE)=5pF
[Table 48] Cio=5pF
C1=C2=2*(10pF – 10pF) = 0pF
[Generic C_stray DON'T USE] C1=C2=2*(10pF – 5pF) = 10pF
[LINK] Calculate C values for each crystal
C1=C2=2*(C_L – C_stray)
Assuming that C_stray = 10pF instead of the "rule of thumb" 5pF
32.768kHz-Y2
32.768kHz
TXC 9H03200031 Load Capacitance C_L = 12.5pF
C5 5pF
C6 5pF
C1=C2=2*(12.5pF – 10pF) = 5pF
[Generic C_stray DON'T USE] C1=C2=2*(12.5pF – 5pF) = 15pF
5V & 3.3V Out
+5V_BATT
+5V_USB
+5V
5V-DiodeArr1
Schottky Diode Array, 5A per, 40V
The STEVE board should not typically pull more than 1.5A at 5V since a connected radio would use ~1A and the rest of the board would use <500mA
3.3V-LinReg1
TLV1117LV33DCYR
VI VO GND
C25 1.0uF
C26 1.0uF
USBC-1
USB_C_Receptacle_16P
+5V_USB
USB C Input
VBUS CC1 CC2 D- D- D+ D+ SBU1 SBU2 SHIELD GND
R11 22
R12 22
R13 5.1k
R14 5.1k
Use USB_C_Receptacle_GCT_USB4105-xx-A_16P_TopMnt_Horizontal Footprint and corresponding GCT connector if available. The HCTL connector is acceptable but not preferred (due to not being on Digikey).
128 Mbit Flash
R15 100k
W25Q128JVSIQ
Flash1
CS CLK DI(IO0) DO(IO1) IO2 IO3 VCC GND
C31 0.1uF
The W25Q128JVSIQ is a SOIC-8 5x5mm part. The W25Q128JVPIQ is a WSON-8 6x5mm part. KiCad has footprints for both
Debug
TagConnect-1
TagConnect_TC2030-NL
VCC RESET SWCLK SWDIO SWO GND
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Size: A4
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Rev:
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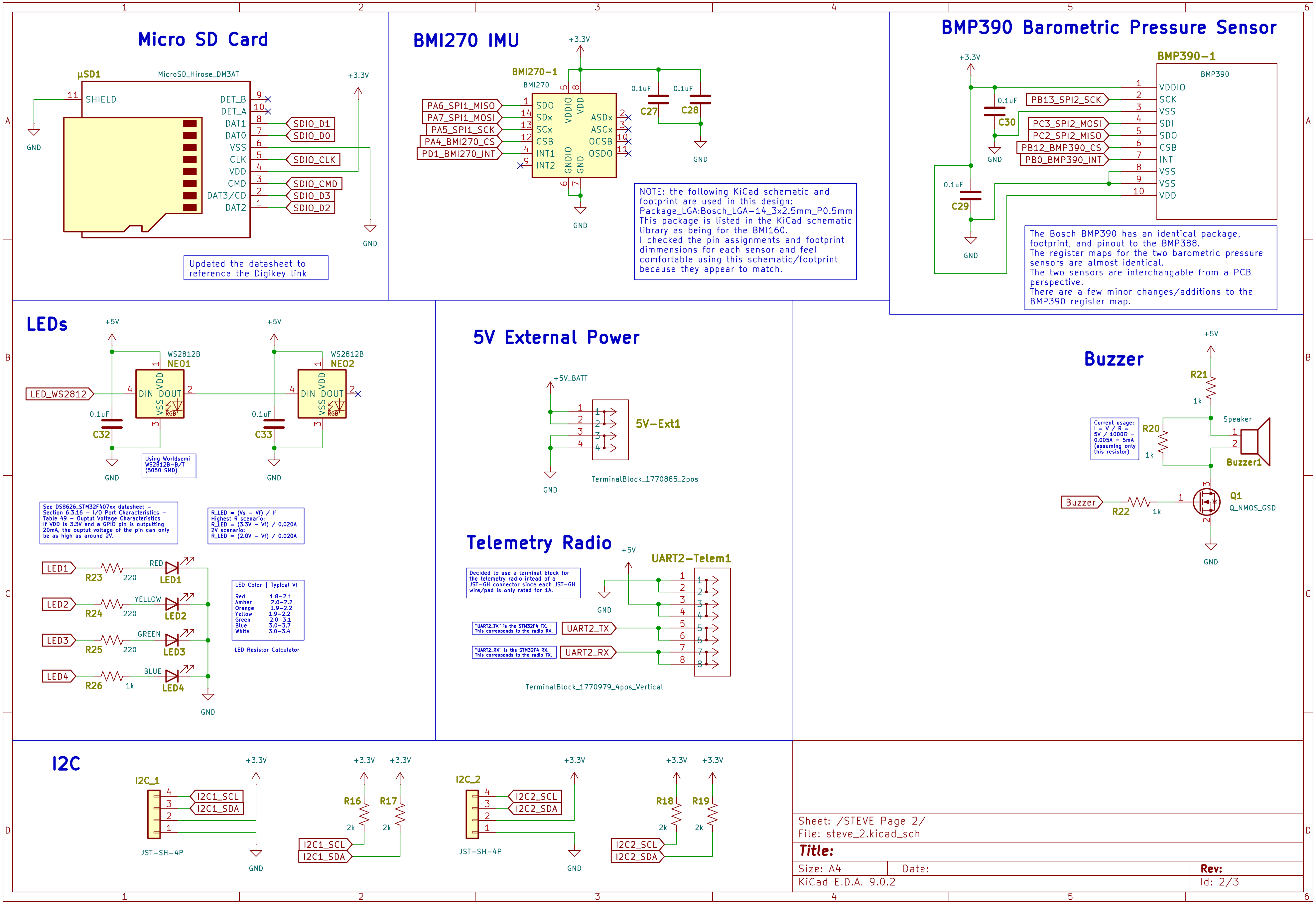
Micro SD Card
µSD1
MicroSD_Hirose_DM3AT
SHIELD
DET_B
DET_A
DAT1
DAT0
VSS
CLK
VDD
CMD
DAT3/CD
DAT2
SDIO_D1
SDIO_D0
SDIO_CLK
SDIO_CMD
SDIO_D3
SDIO_D2
+3.3V
GND
Updated the datasheet to reference the Digikey link
BMI270 IMU
BMI270-1
BMI270
PA6_SPI1_MISO
PA7_SPI1_MOSI
PA5_SPI1_SCK
PA4_BMI270_CS
PD1_BMI270_INT
SDO
SDx
SCx
CSB
INT1
INT2
VDDIO
VDD
GNDIO
GND
ASDx
ASCx
OCSB
OSDO
0.1uF
C27
C28
NOTE: the following KiCad schematic and footprint are used in this design:
Package_LGA:Bosch_LGA-14_3x2.5mm_P0.5mm
This package is listed in the KiCad schematic library as being for the BMI160.
I checked the pin assignments and footprint dimmensions for each sensor and feel comfortable using this schematic/footprint because they appear to match.
BMP390 Barometric Pressure Sensor
BMP390-1
BMP390
PB13_SPI2_SCK
PC3_SPI2_MOSI
PC2_SPI2_MISO
PB12_BMP390_CS
PB0_BMP390_INT
VDDIO
SCK
VSS
SDI
SDO
CSB
INT
VDD
C29
C30
The Bosch BMP390 has an identical package, footprint, and pinout to the BMP388.
The register maps for the two barometric pressure sensors are almost identical.
The two sensors are interchangable from a PCB perspective.
There are a few minor changes/additions to the BMP390 register map.
LEDs
+5V
WS2812B
NEO1
NEO2
LED_WS2812
DIN
DOUT
VDD
VSS
RGB
C32
C33
Using Worldsemi WS2812B-B/T (5050 SMD)
See DS8626_STM32F407xx datasheet - Section 6.3.16 - I/O Port Characteristics - Table 49 - Ouptut Voltage Characteristics
If VDD is 3.3V and a GPIO pin is outputting 20mA, the ouptut voltage of the pin can only be as high as around 2V.
R_LED = (Vs - Vf) / If
Highest R scenario:
R_LED = (3.3V - Vf) / 0.020A
2V scenario:
R_LED = (2.0V - Vf) / 0.020A
LED1 R23 220 RED LED1
LED2 R24 220 YELLOW LED2
LED3 R25 220 GREEN LED3
LED4 R26 1k BLUE LED4
LED Color | Typical Vf
Red 1.8-2.1
Amber 2.0-2.2
Orange 1.9-2.2
Yellow 1.9-2.2
Green 2.0-3.1
Blue 3.0-3.7
White 3.0-3.4
LED Resistor Calculator
5V External Power
+5V_BATT
5V-Ext1
TerminalBlock_1770885_2pos
Telemetry Radio
Decided to use a terminal block for the telemetry radio intead of a JST-GH connector since each JST-GH wire/pad is only rated for 1A.
UART2-Telem1
"UART2_TX" is the STM32F4 TX. This corresponds to the radio RX.
"UART2_RX" is the STM32F4 RX. This corresponds to the radio TX.
UART2_TX
UART2_RX
TerminalBlock_1770979_4pos_Vertical
Buzzer
R21 1k
Speaker
Buzzer1
Current usage: I = V / R = 5V / 1000Ω = 0.005A = 5mA (assuming only this resistor)
R20 1k
Buzzer
R22 1k
Q1
Q_NMOS_GSD
I2C
I2C_1
I2C1_SCL
I2C1_SDA
JST-SH-4P
R16 2k
R17 2k
I2C_2
I2C2_SCL
I2C2_SDA
R18 2k
R19 2k
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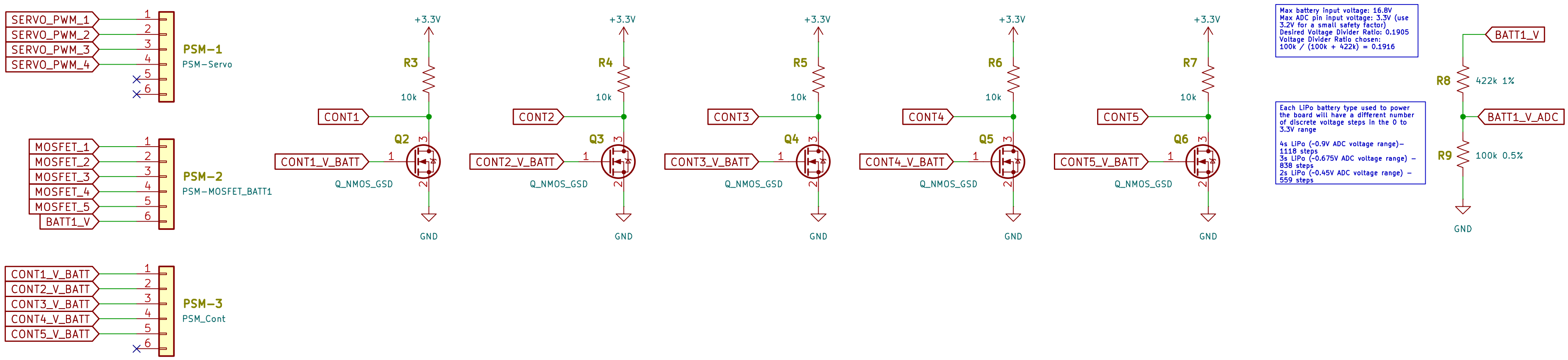
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.2
Id: 2/3

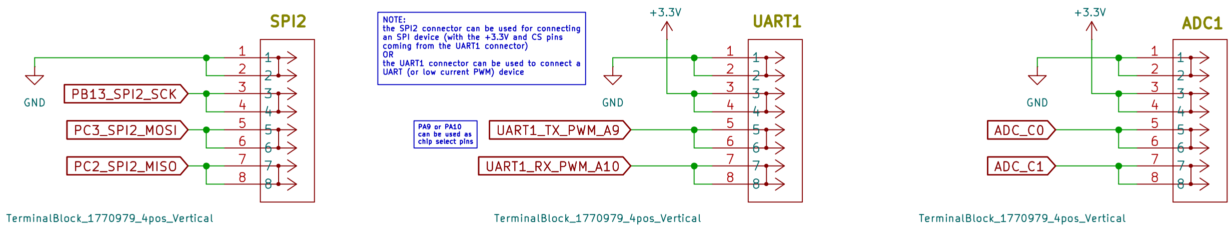
Servo, Mosfet, and Battery Signals

NOTE:
When a CONT# signal is grounded, there is continuity

NOTE: The Nexperia 2N7002BK-215 Mosfet has a max V_GS of 20V.
A voltage divider before the mosfet or a different mosfet would need to be used if the battery voltage is more than 20V.



Additional I/O



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